The Exhaustion Paradox:

How Fragmented Timetables Drain Student Energy

IDHV Assignment 1

Introduction to Data Handling and Visualization

Name: Vignesh Balamurugan M.B

Roll Number: S20230030422

Branch: AI & DS 3rd Year

Table of Contents

1. Introduction

1.1 The Research Problem
1.2 Context and Motivation
1.3 Personal Inspiration

2. Methodology

2.1 Defining the Study
2.2 Crafting the Survey
2.3 Variable Dictionary
2.4 Who Took the Survey?
2.5 Gathering the Data

3. Data Preprocessing

3.1 The Reality of Messy Data
3.2 Untangling Text and Blank Cells
3.3 The Phantom Outlier
3.4 Scaling the Variables
3.5 Feature Engineering

4. Analysis and Findings

4.1 A Bleak Baseline
4.2 Dissecting the Distributions
4.3 The Correlation Heatmap
4.4 Proving the Fatigue Trend
4.5 The Illusion of Rest (Location Impact)
4.6 Gender Dynamics and Logistics
4.7 The Hostel Trip Penalty
4.8 Branch Nuances
4.9 The Verdict from Students
4.10 Time Wasted

5. Conclusion

6. Appendix — Questionnaire Screenshots

1. Introduction

1.1 The Research Problem

Does having "free time" between classes actually leave you feeling rested, or does it quietly drain your energy instead? That question is the whole basis of this study. I wanted to look specifically at how a fragmented, gap-heavy timetable affects how exhausted a student feels by the end of the day. The real issue worth investigating was whether short, isolated windows of free time between lectures actually help students recover and stay productive — or whether they just turn the campus into an unusually tiring waiting room.

1.2 Context and Motivation

There is a logistical reality at IIIT Sri City that makes this question more pressing than it sounds. The academic block sits roughly a kilometre away from the hostels, and every trip between them — in Andhra Pradesh heat — costs you something physically. In first and second year, the schedule was shared across the batch. Everyone had the same block timetable, which meant a certain predictable rhythm to the day. By third year, that structure disappears entirely. Elective choices mean everyone ends up with a different, personalised schedule, and for most people, those schedules are full of holes. Because we all pick different subjects, the previously synchronized timetables break apart into individualized, fragmented puzzles that vary wildly from student to student.

1.3 Personal Inspiration

I will be honest — this project started from frustration, not curiosity. The specific scenario that pushed me to actually do something about it: arriving for a 9:45 AM lecture, finishing it, and then discovering that the next class was not until noon. That gap is a problem. It is too short to make a real trip back to the hostel worth it — you cannot actually sleep and recover in that window. But it is also too long to just stand around in a corridor doing nothing. The result was always the same: sitting in the library or at Bhagat Ji, going through the motions of productivity, achieving essentially nothing, and somehow feeling completely drained by the time the next class rolled around.

On the other hand, a three-hour gap is a completely different situation. That is actually long enough to walk back, eat properly, get a real nap in, and come back to the academic block feeling like a functioning person again. My suspicion, going into this, was that it is specifically the volume of those awkward one-hour gaps — not the longer breaks — that is responsible for the zombie-like state most third-years settle into by 6 PM. I decided it was time to stop hypothesising and actually test it with data.

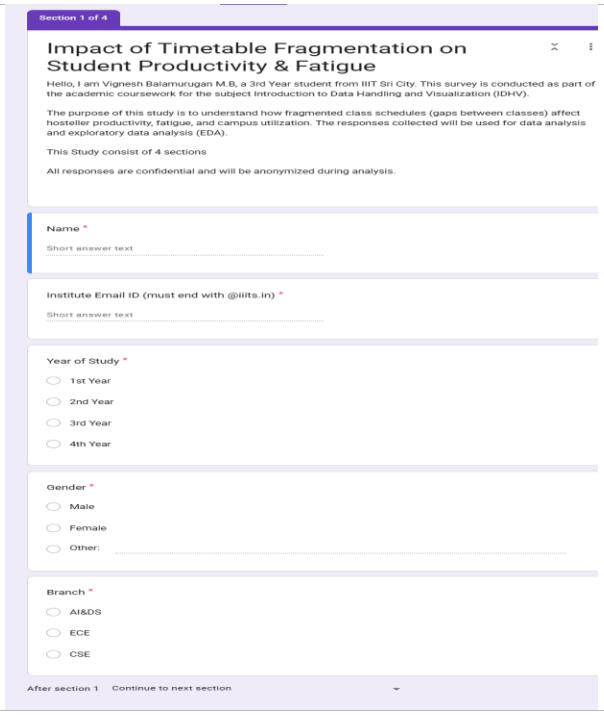
2. Methodology

2.1 Defining the Study

Just asking classmates whether they feel tired was never going to work — everyone always says yes, which tells you nothing. The goal was to actually quantify which specific factors of a student's timetable are driving that exhaustion. So I designed this as a cross-sectional survey, but with a deliberate scope: responses would only cover the two most demanding days of each student's week. Including lighter days in the analysis would just dilute the signal. The aim was to capture data from the genuinely difficult days — the ones where schedule fragmentation is actively causing fatigue and psychological wear, not the days where students coast through two afternoon lectures.

2.2 Crafting the Survey

The Google Form was kept deliberately short: 19 targeted questions split across four logical sections. That brevity was intentional. Longer surveys get filled in carelessly or abandoned halfway through, and I needed clean, usable data. The question format was a deliberate mix of categorical dropdowns and numerical linear scales. That combination was the minimum required to generate a dataset rich enough to run a proper Exploratory Data Analysis (EDA) without it falling apart when I started grouping and correlating things in Pandas.


Section 1 of 4

Impact of Timetable Fragmentation on Student Productivity & Fatigue

Hello, I am Vignesh Balamurugan M.B, a 3rd Year student from IIIT Sri City. This survey is conducted as part of the academic coursework for the subject Introduction to Data Handling and Visualization (IDHV).

The purpose of this study is to understand how fragmented class schedules (gaps between classes) affect hosteller productivity, fatigue, and campus utilization. The responses collected will be used for data analysis and exploratory data analysis (EDA).

This Study consist of 4 sections

All responses are confidential and will be anonymized during analysis.

Name *

Short answer text

Institute Email ID (must end with @iiits.in) *

Short answer text

Year of Study *

- 1st Year
- 2nd Year
- 3rd Year
- 4th Year

Gender *

- Male
- Female
- Other:

Branch *

- AI&DS
- ECE
- CSE

After section 1 Continue to next section

2.3 Variable Dictionary

Before writing a single line of analysis code, I mapped out exactly what each column in the dataset represented. Keeping that dictionary clear from the start was how I stayed sane during the Pandas work later:

- Authentication — Name and Email, used to verify respondents were part of the IIIT Sri City community.
- Demographics — Branch (CSE, ECE, or AI&DS) and Gender.

- The Load — Total number of classes scheduled on each student's two busiest days.
- The Fragmentation — Single Gaps (exact 1-hour holes in the schedule) and Double Gaps (2-hour holes).
- The Refuge — Short Gap Location: where students actually spent those gap periods — the Library, an AC Classroom, back in the Hostel, or the Canteen.
- The Journey — Hostel Trips: the number of times a student made the walk back and forth to the residential blocks on those days.
- The Cost — Productivity (a self-rated 1–10 score for how much was actually accomplished during gaps) and Fatigue (an honest 1–10 score for end-of-day exhaustion).

2.4 Who Took the Survey?

The survey link was shared exclusively with 3rd Year students in Common Whatsapp group . To block any spam entries — because I know for a fact that certain friends would have happily submitted nonsense responses to sabotage the data — the form required a valid @iiits.in institutional email address to submit. That filter worked. I ended up with exactly 60 valid responses. The sample skewed heavily toward CSE, which contributed 39 respondents, with AI&DS adding 12 and ECE adding 9. The gender breakdown was 48 males and 12 females, which is a ratio that closely mirrors the actual composition of our college.

2.5 Gathering the Data

Data collection wrapped up in done in March 2026. One of the few advantages of using Google Forms is that the export to CSV is immediate, clean, and comes with automatic timestamps. However, opening the raw spreadsheet for the first time was a reminder that people are genuinely terrible at following straightforward instructions. Despite the form clearly asking for numeric values, several respondents had typed full English words into the boxes instead. That, combined with some partially incomplete submissions, meant the data needed real work before any analysis could happen.

3. Data Preprocessing

3.1 The Reality of Messy Data

Loading a 60-row CSV into Pandas sounds straightforward. It was not. The dataframe had problems in multiple columns. Despite the form explicitly requesting numeric counts, a handful of respondents had typed things like "seven classes" or "eight" into fields that were supposed to contain just a number. On top of that, some respondents had clearly stopped filling the form out partway through, which left a scattering of NaN values spread across several rows — not concentrated in one place, just random gaps that needed handling before any groupby or correlation operation could run without throwing errors.

3.2 Untangling Text and Blank Cells

The first thing I tackled was the free-text numeric fields. I wrote a regex script to extract integers from those rogue string entries. That handled most cases, but for the ones that had written out numbers in full English, I backed the regex up with a small custom dictionary — so "eight" mapped to 8, "seven" to 7, and so on. For the categorical columns, capitalisation inconsistencies ("yes" in one row, "No" in the next) were standardised using .str.lower() followed by .str.capitalize(), which prevented the pandas groupby operations from treating identical answers as different categories.

For the five blank cells scattered across the dataset, dropping the rows entirely was not an option. My working sample was already at 60, and losing rows would risk bringing the usable count too close to the minimum of 50 that the assignment parameters required. Instead, I applied median imputation. The choice of median over mean was deliberate: several numeric columns had extreme values lurking in them, and using the mean for imputation would have let those outliers pull the filled-in values away from the realistic range for that variable.

3.3 The Phantom Outlier

Running an IQR (Interquartile Range) outlier check on the numeric columns surfaced something immediately. One respondent had reported walking back to their hostel 15 times over the span of two days. Given the actual slot structure of our timetable and the physical constraints of the day, that number is not just high — it is logistically impossible. The IQR algorithm correctly identified it as an outlier, and I replaced the rogue value of 15 with the column median of 2, which is a far more realistic figure.

3.4 Scaling the Variables

Before generating the correlation heatmap, I ran seven numeric columns through a Min-Max Scaler to compress all of them into a uniform 0-to-1 range. Without that step, a variable like the fatigue score — which spans 1 to 10 — and a variable like single gap count — which might range from 0 to 4 — would produce wildly uneven colour distributions in the heatmap. The wider-range column would dominate the visual, making the tighter-range columns look like they have no variation at all. Scaling fixed that before it became a problem.

3.5 Feature Engineering

Rather than limiting the analysis to the raw survey columns alone, I constructed three additional compound metrics to capture relationships that the individual variables could not express on their own:

- Total Gap Score — A weighted sum: Long Gaps multiplied by 3, Double Gaps by 2, Single Gaps by 1. The weighting reflects the increasing structural disruption each gap type causes. The resulting score gives a single number that summarises how fragmented a student's day is, overall.
- Efficiency Ratio — Productivity score divided by Total Gap Score. This asks a pointed question: given how broken-up their day was, are students actually making use of that time, or just enduring it?

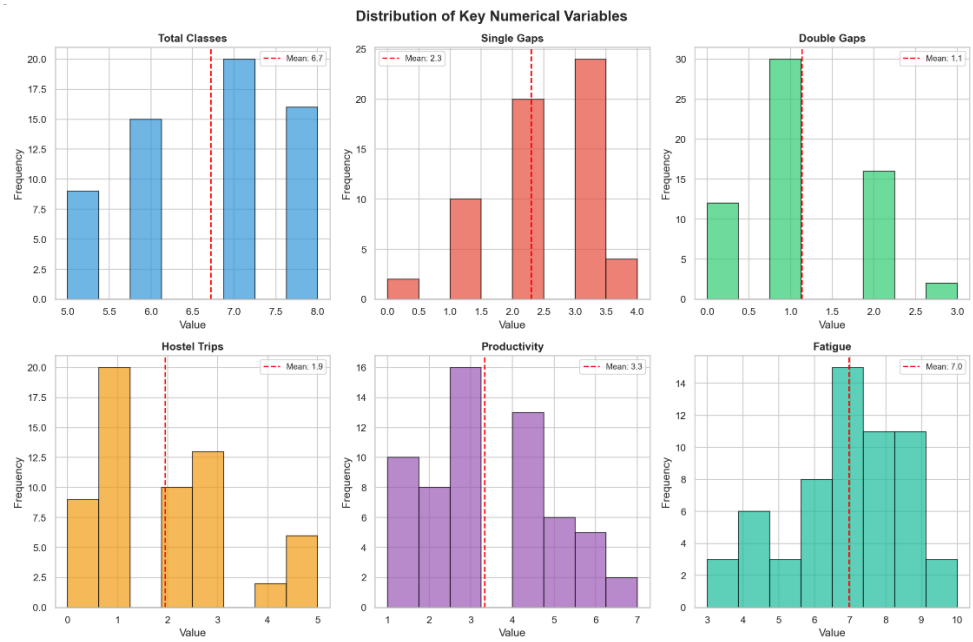
- Fatigue Cost — Exhaustion score divided by Productivity score. A high value here means the student paid a heavy physical and mental price while producing almost no academic output. The worst possible outcome.

4. Analysis and Findings

4.1 A Bleak Baseline

Running .describe() on the cleaned dataset produced a summary that was not particularly encouraging to look at. The mean fatigue score came in just below 7 out of 10. That is a high number — students, on their two most demanding days of the week, are genuinely and consistently exhausted. At the same time, the average productivity score recorded during gap periods was 3.3 out of 10. That is the part that really stings. Students are absorbing the full physical cost of spending an entire day on campus in the heat, making multiple trips between buildings, sitting through back-to-back waiting periods — and at the end of it, the self-reported academic output during those gaps is barely above the floor. The negative skew of -0.51 on the fatigue distribution made this even clearer: responses were not spread evenly across the 1–10 range. The overwhelming majority of the cohort clustered toward the high end of the exhaustion scale.

4.2 Dissecting the Distributions

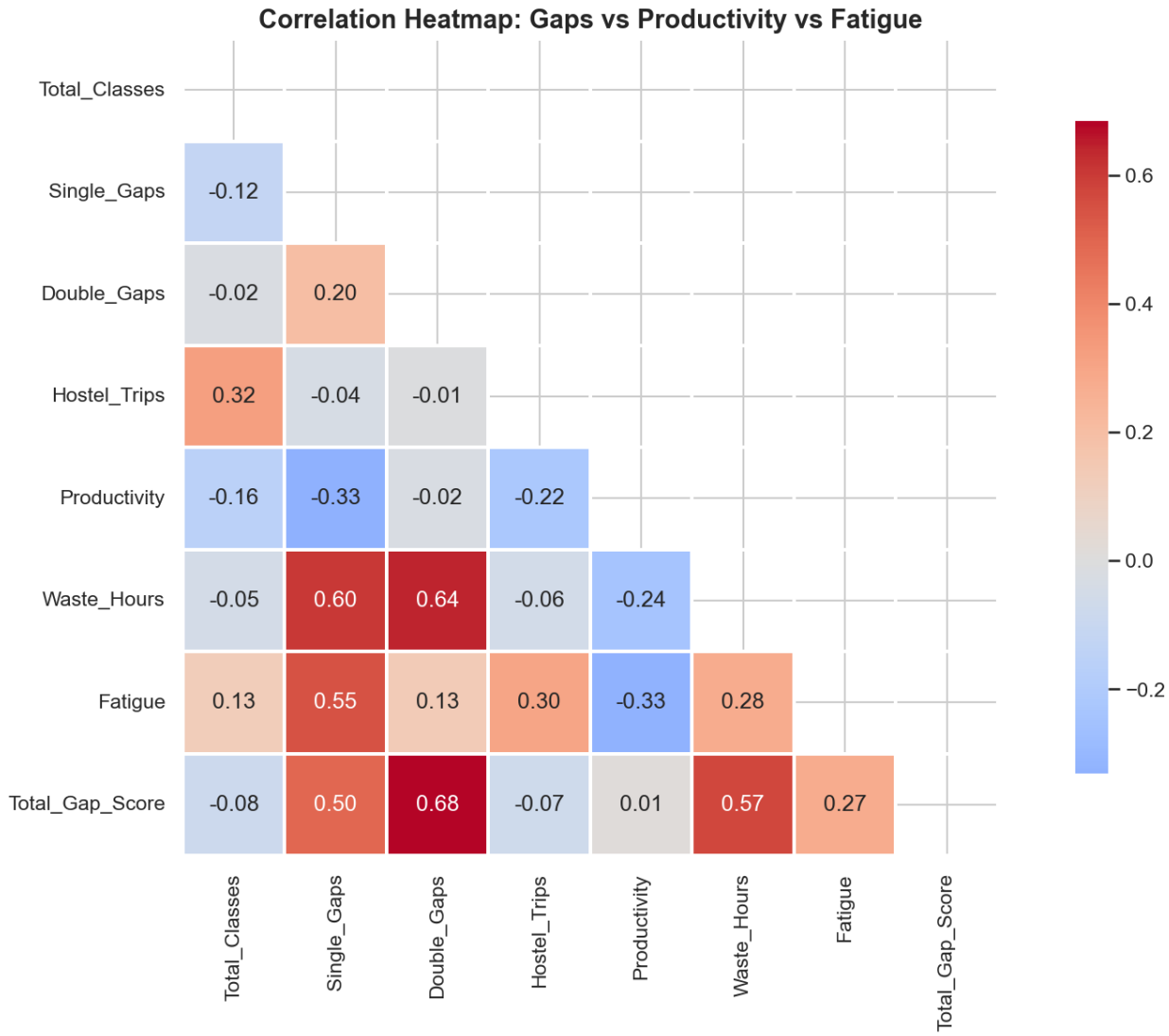


plot1_distributions.png

The histograms added visual detail to what the summary statistics had already suggested. Total class counts were fairly evenly spread across the 5-to-8 range, which means differences in fatigue are not simply explained by some students having twice as many classes as others — the load is roughly comparable. The single gaps histogram was more dramatic. It spiked hard at 2 and 3 — meaning nearly every person in the sample is dealing with at least two or three isolated one-hour holes in their schedule on their worst days. The wasted hours chart, by contrast, was all over the place, spread across a wide range with no clear concentration. That tells you something: while

everyone wastes time during those gaps, the volume varies a lot from person to person, which is worth keeping in mind when interpreting the productivity scores.

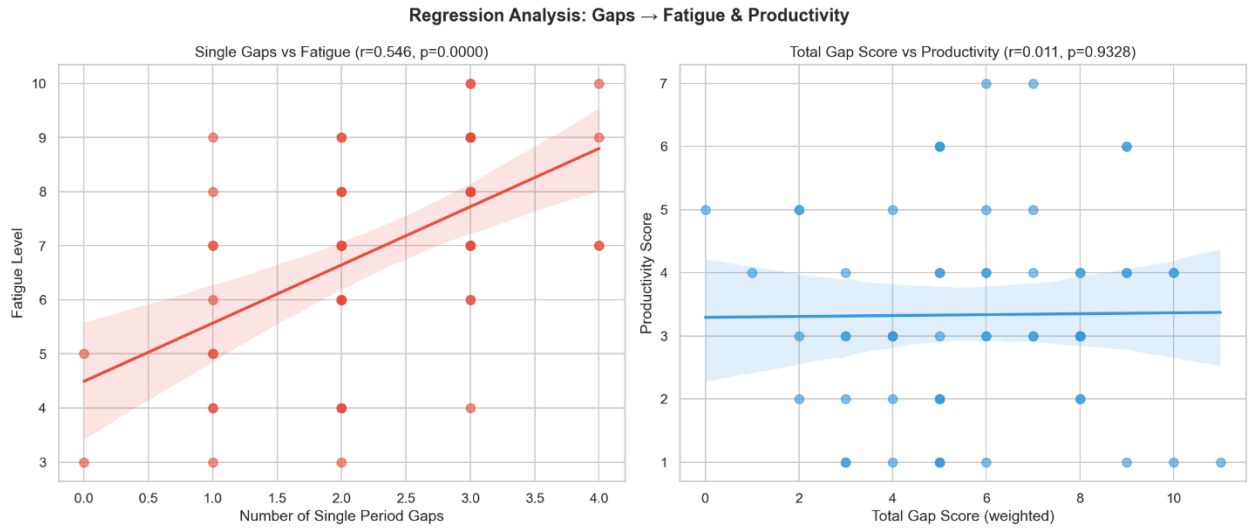
4.3 The Correlation Heatmap



plot2_correlation_heatmap.png

The Pearson correlation heatmap was the most important output in this entire project. The two numbers that stood out most clearly: the correlation between Single Gaps and Fatigue was 0.55, with a p-value low enough to treat as essentially zero. That is real, statistically defensible evidence that one-hour gaps are a driver of end-of-day exhaustion, not just an incidental feature of busy schedules. Just as telling was the 0.60 correlation between Single Gaps and Wasted Hours. Students are not spending those gaps productively in the library — they are largely sitting around, waiting, and accomplishing very little. The heatmap confirmed both of those relationships visually through dark, saturated squares that made the patterns hard to dismiss.

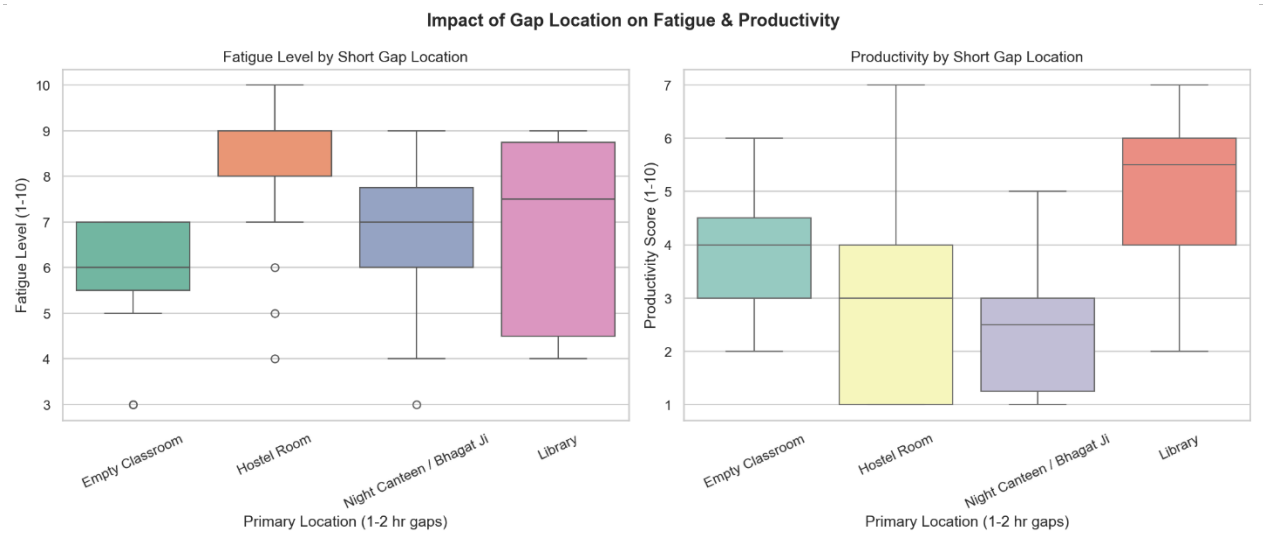
4.4 Proving the Fatigue Trend



plot4_regression_analysis.png

The seaborn regression plot made the relationship between single gaps and fatigue even harder to argue with. The slope of the regression line is steep: the model estimates that each additional one-hour gap in a student's day is associated with roughly a one-point increase in their end-of-day fatigue score on the 10-point scale. That is a meaningful effect per gap. The companion scatter plot — mapping total gap count against productivity — told a different story. That line is essentially flat. Fragmentation alone does not directly kill your productivity rating; how you choose to spend those fragmented periods is what determines whether you get anything done. The gaps create the conditions for wasted time, but the student still makes the choice.

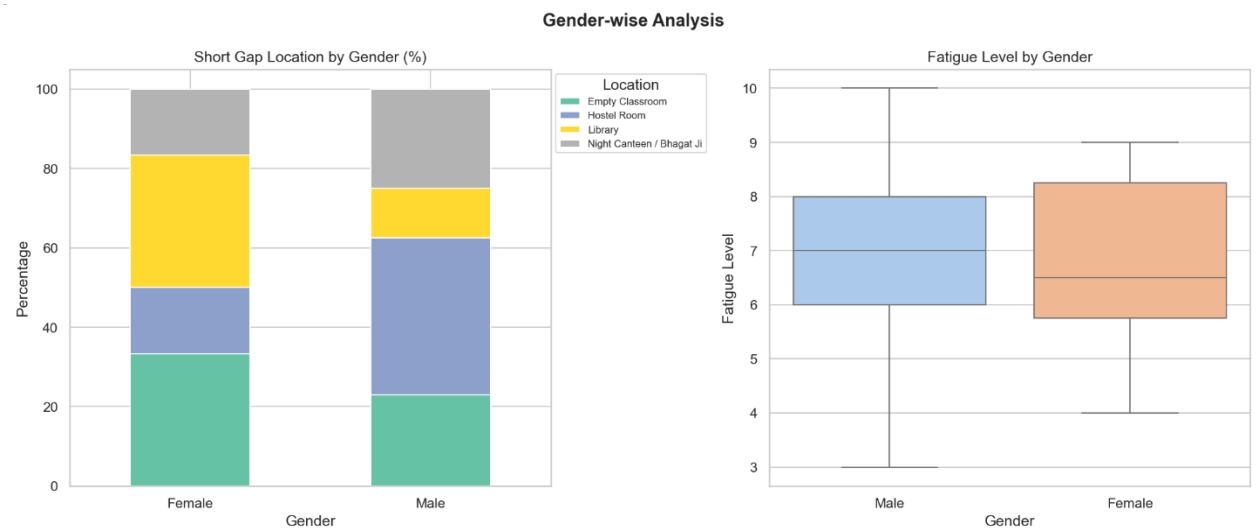
4.5 The Illusion of Rest (Location Impact)



plot3_location_boxplots.png

This visualisation was the one that upended my assumptions the most. Going in, I had expected students who went back to their hostel rooms during gaps to show lower fatigue scores — because obviously a bed is more restful than a library chair. The data disagreed. Students who trekked back to their hostel rooms during gaps reported the highest average fatigue in the entire cohort, sitting at 8.0 out of 10. The explanation, looking at it after the fact, is fairly straightforward: the walk itself — roughly a kilometre each way, in the afternoon heat — costs more in terms of physical exertion than whatever recovery the hostel room provides. You are not resting; you are trading one form of strain for another. The group with the lowest average fatigue was the students who stayed put in empty, air-conditioned classrooms, who reported an average of 5.9. They did not gain anything special by doing so. They just avoided the walk, and that alone was enough to make a noticeable difference.

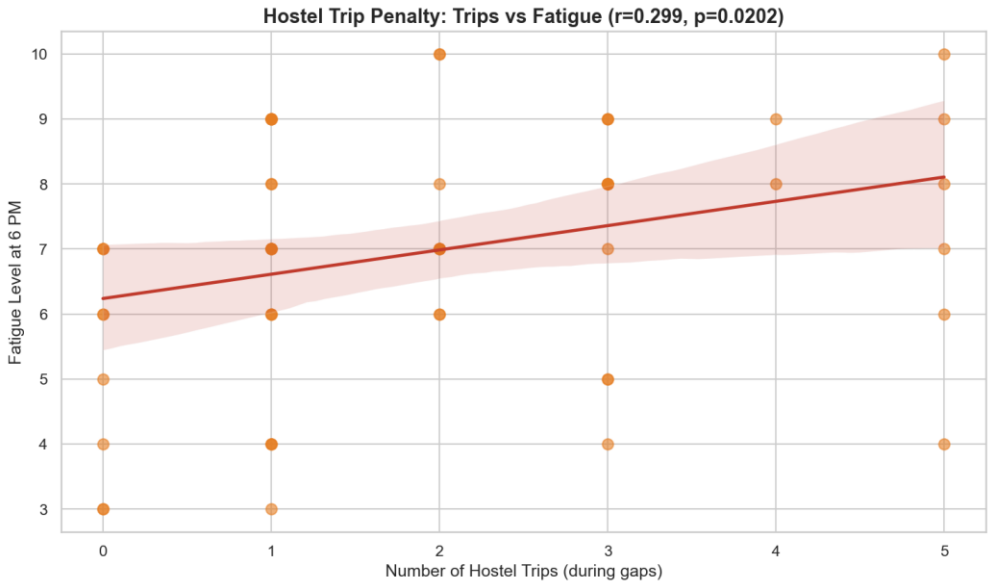
4.6 Gender Dynamics and Logistics



plot5_gender_analysis.png

The stacked bar chart revealed a logistical divide that is entirely structural rather than personal. Because the women's hostel at IIIT Sri City requires a dedicated bus rather than just a walk, making a round trip during a one-hour gap is not a realistic option for female students — the logistics simply do not allow it. The data reflected this: 80% of female respondents spent their gap periods in the library or in classrooms. Male students, whose hostels are accessible on foot even if it is not pleasant, made the attempt to go back roughly 40% of the time. Yet despite those very different coping strategies, the end-of-day fatigue scores were nearly identical across the two groups: 7.0 for males and 6.7 for females. That is a narrow enough gap to suggest that the specific location chosen during gaps matters significantly less than the act of making the walk in the first place.

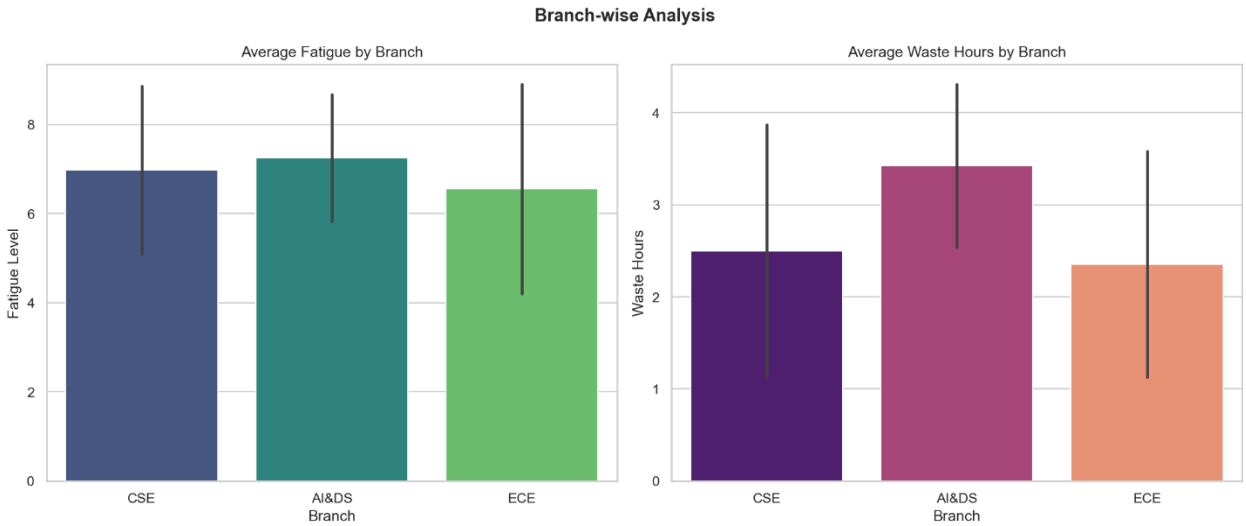
4.7 The Hostel Trip Penalty



plot7_hostel_trip_penalty.png

If the location boxplot raised the question, this scatter plot answered it directly. Each additional round trip from the academic block to the residential hostels and back is associated with approximately a 0.4-point increase in overall fatigue score. It is not a dramatic number in isolation, but when a student makes two or three of those trips across a day, the cumulative effect is real and measurable. The straightforward implication is that staying within the academic block, even in a less comfortable setting, is the only strategy that does not actively make the day harder.

4.8 Branch Nuances

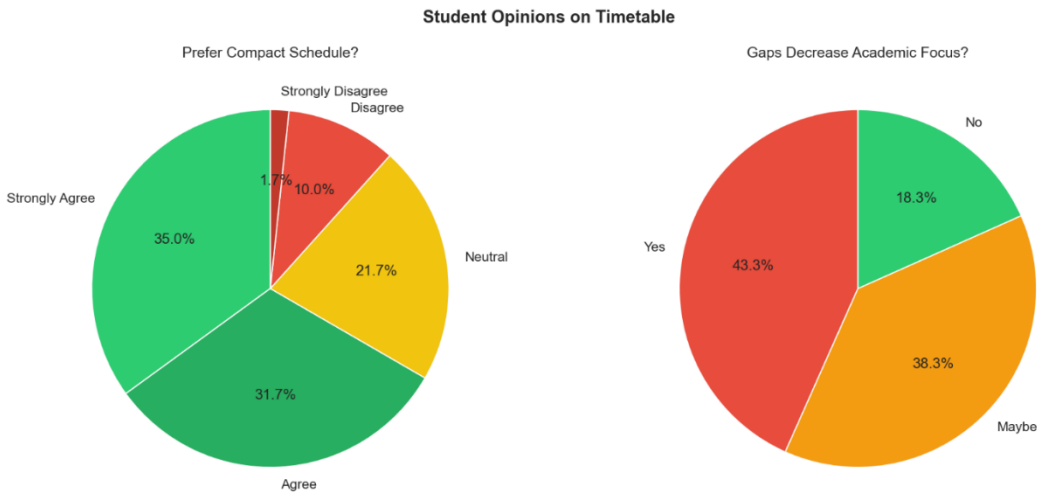


plot6_branch_analysis.png

Comparing fatigue across the three represented branches produced some variation. CSE and ECE students came in slightly higher on the exhaustion index than their AI&DS counterparts, which could plausibly be explained by lab slots — longer fixed-duration sessions reduce scheduling flexibility and change the overall texture of the day. However, I am cautious about how much weight to put on this observation. The ECE subsample had only 9 respondents, and AI&DS had

12. Drawing firm conclusions about branch-specific differences from sample sizes that small feels statistically precarious, and any such conclusion would need a much larger dataset to be credible.

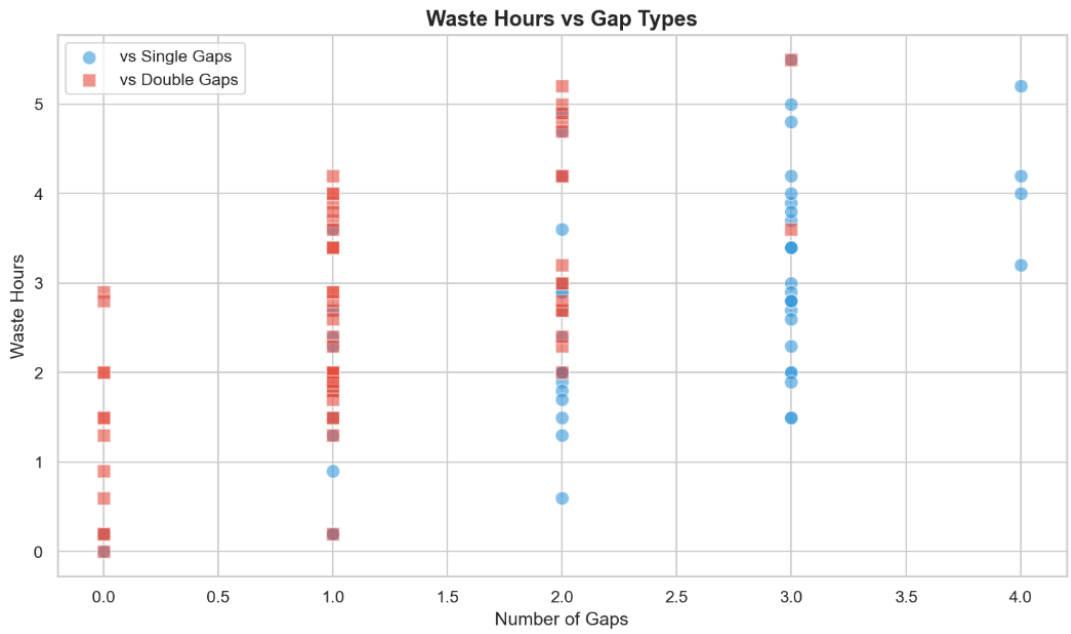
4.9 The Verdict from Students



plot8_opinions.png

When asked directly what they would prefer, 67% of respondents said they would choose a compressed, back-to-back class schedule over the current fragmented one. The opposition was minimal. There was no meaningful split of opinion here — the distaste for the one-hour-gap-riddled timetable was close to universal among the students who responded. It is worth noting that this preference is expressed despite the fact that a condensed schedule would mean fewer breaks, less time between classes, and a more intense stretch of back-to-back lectures. Students are still choosing that over the current arrangement, which tells you something about how genuinely draining the fragmentation is.

4.10 Time Wasted



plot9_waste_vs_gaps.png

Plotting wasted hours against both single and double gaps produced an interesting result. The distributions overlapped substantially, meaning that a two-hour gap does not lead to significantly better use of time than a one-hour gap — it just provides a longer window to do nothing productive. A two-hour gap is not double the productivity opportunity; it appears to be roughly double the opportunity to end up on YouTube. The takeaway is that gap duration beyond a certain threshold — probably something closer to three hours, where a genuine hostel-rest-return cycle becomes feasible — does not linearly improve how students use their time. The waste scales with duration in the absence of a structural reason to actually do something.

5. Conclusion

Working through these 60 rows of data has done a pretty effective job of dismantling the comfortable assumption that free time between classes is a net positive. The evidence from this study points in a consistent direction: one-hour gaps, which are by far the most common type of gap in fragmented third-year schedules, are not restful. They are draining. The correlation of 0.546 between single gaps and end-of-day fatigue is clear enough to state with confidence that every additional isolated hour embedded in a student's timetable is pushing their exhaustion higher by the end of the day.

The hostel finding added another layer to this. The instinct to walk back to the room during a gap — the assumption that a bed equals recovery — turns out to be a trap. The walk to and from the hostels under the sun costs more than the rest gives back. Students who stayed in air-conditioned classrooms fared meaningfully better on fatigue, not because they were doing anything particularly effective, but because they avoided inflicting additional physical strain on themselves.

The student preference data underlines all of this. The majority of respondents would trade their scattered mid-day gaps for a tighter, more compressed schedule where classes run back-to-back. The intuition seems to be: get through it faster, and actually be free at the end of it, rather than technically having "breaks" that cost more than they recover.

That said, there are real limitations in this analysis that have to be acknowledged. The most significant is that both fatigue and productivity were self-reported on a 1–10 scale — which means there is no calibration across respondents. One person’s 8 might genuinely be another person’s 5, depending on baseline tolerance and framing. The study also restricts itself entirely to third-year students, which means it says nothing about how freshmen or second-years experience their more rigid, synchronised timetables. A comparison across years would have been informative. On top of that, variables like the ambient temperature on specific days, whether AC was actually functional in a given space, and individual caffeine or sleep habits were all left out of scope entirely — each of those could plausibly interact with fatigue in ways this dataset cannot account for. Any follow-up study would need to address at least some of these gaps to make more robust claims.

6. Appendix — QUESTIONARIES

Below are the screenshots of the Google Form that was used to collect the data for this study. The form was structured across four sequential sections, covering a total of 19 questions designed to capture schedule structure, gap behaviour, location choices, and self-assessed fatigue and productivity outcomes.

QUESTIONS :

Section 1: Student Profile

This section confirms the respondent is a valid IIIT Sri City student and gathers basic demographic data.

1. **Name** (Short Answer Text) - *Required*
2. **Institute Email ID** (Short Answer Text) - *Required* (Used to verify @iiits.in domain)
3. **Year of Study** (Multiple Choice: 1st, 2nd, 3rd, 4th) - *Required*
4. **Gender** (Multiple Choice: Male, Female, Prefer not to say) - *Required*
5. **Branch** (Multiple Choice: CSE, ECE, AI&DS) – *Required*

Section 2: Timetable Structure

This section captures the exact structural fragmentation of their schedule during their most exhausted days.

6. **Think about your 2 busiest/most packed days of the week. How many total class slots do you roughly attend across BOTH of those days combined?** (Dropdown: 4, 5, 6, 7, 8, 9, 10) – *Required*
7. **On those 2 busy days, how many SINGLE PERIOD (1-hour) gaps do you have in total?** (Linear Scale: 0 to 5+) - *Required*
8. **On those 2 busy days, how many DOUBLE PERIOD (2-hour) gaps do you have in total?** (Linear Scale: 0 to 5+) - *Required*
9. **Do you have any LONG gaps (3+ hours) on those days?** (Multiple Choice: Yes, No) – *Required*
10. **If Yes, how many long gaps in total?** (Linear Scale: 1 to 3+) – *Optional*

Section 3: Gap Utilization & Logistics

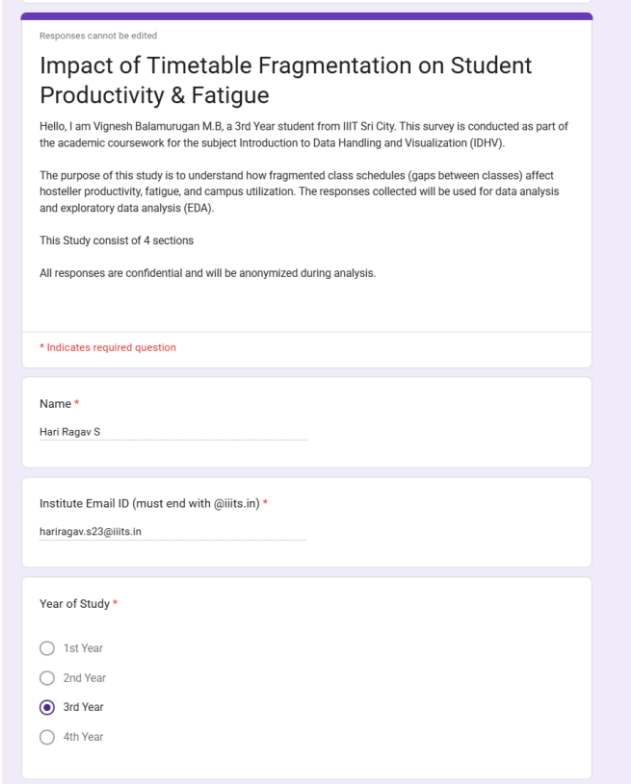
This section tracks where students go during their free time and the physical toll of navigating the campus.

11. **Where do you mostly spend your SHORT gaps (1-2 hours)?** (Multiple Choice: Library, Night Canteen / Bhagat Ji, Empty AC Classroom, Hostel Room) - *Required*
12. **Where do you mostly spend your LONG gaps (3+ hours)?** (Multiple Choice: Library, Night Canteen / Bhagat Ji, Empty AC Classroom, Hostel Room) - *Optional*
13. **Over those 2 busy days, how many times do you walk BACK to your hostel room during the day (excluding the final trip at night)?** (Dropdown: 0, 1, 2, 3, 4, 5+) - *Required*
14. **What do you usually end up doing during these gaps?** (Checkboxes: Studying/Assignments, Scrolling Phone/Social Media, Sleeping/Resting, Talking with Friends) – *Required*

Section 4: Productivity and Fatigue

This section collects the core metrics of the study: self-reported academic output and physical exhaustion.

15. **On a scale of 1 to 10, how PRODUCTIVE do you feel during your gap hours?** (Linear Scale: 1 = Complete waste of time, 10 = Highly productive) – *Required*
16. **Roughly how many hours do you feel are completely WASTED (neither studying nor properly resting) on those 2 busy days combined?** (Short Answer Text - Number) - *Required*
17. **On those busy days, how FATIGUED/TIRED do you feel by 6:00 PM?** (Linear Scale: 1 = Totally energetic, 10 = Completely dead/exhausted) - *Required*
18. **Do you feel that having random gaps between classes decreases your focus for the next class?** (Multiple Choice: Yes, No, Maybe) - *Required*
19. **Would you prefer a more compact schedule (back-to-back classes) if it meant finishing earlier in the day?** (Multiple Choice: Strongly Agree, Agree, Neutral, Disagree, Strongly Disagree) - *Required*



Responses cannot be edited

Impact of Timetable Fragmentation on Student Productivity & Fatigue

Hello, I am Vignesh Balamurugan M.B, a 3rd Year student from IIIT Sri City. This survey is conducted as part of the academic coursework for the subject Introduction to Data Handling and Visualization (IDHV).

The purpose of this study is to understand how fragmented class schedules (gaps between classes) affect hosteller productivity, fatigue, and campus utilization. The responses collected will be used for data analysis and exploratory data analysis (EDA).

This Study consist of 4 sections

All responses are confidential and will be anonymized during analysis.

* Indicates required question

Name *

Hari Ragav S

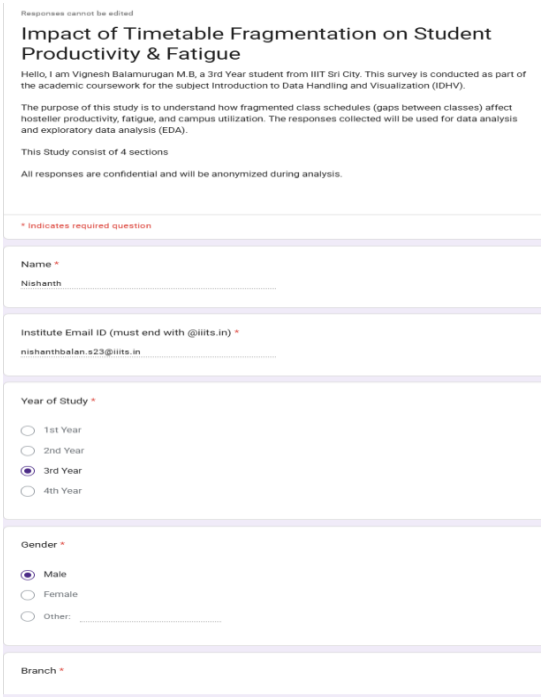
Institute Email ID (must end with @iiits.in) *

hariragav.s23@iiits.in

Year of Study *

- 1st Year
- 2nd Year
- 3rd Year
- 4th Year

Example response from an individual



Responses cannot be edited

Impact of Timetable Fragmentation on Student Productivity & Fatigue

Hello, I am Vignesh Balamurugan M.B, a 3rd Year student from IIIT Sri City. This survey is conducted as part of the academic coursework for the subject Introduction to Data Handling and Visualization (IDHV).

The purpose of this study is to understand how fragmented class schedules (gaps between classes) affect hosteller productivity, fatigue, and campus utilization. The responses collected will be used for data analysis and exploratory data analysis (EDA).

This Study consist of 4 sections

All responses are confidential and will be anonymized during analysis.

* Indicates required question

Name *

Nishanth

Institute Email ID (must end with @iiits.in) *

nishanthbalan.s23@iiits.in

Year of Study *

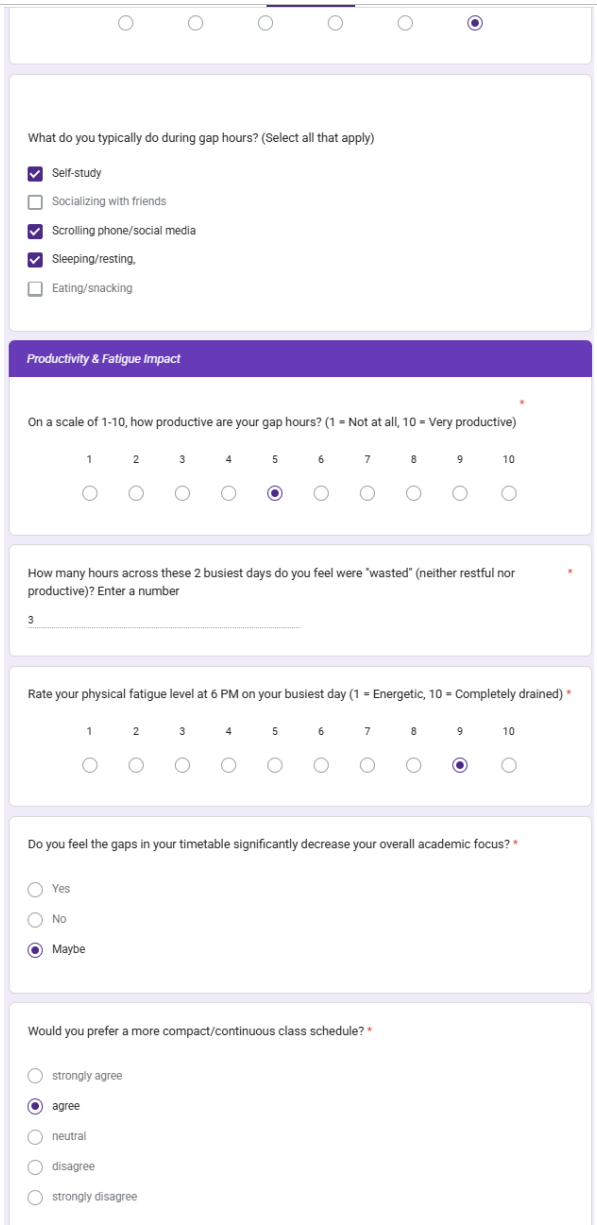
- 1st Year
- 2nd Year
- 3rd Year
- 4th Year

Gender *

- Male
- Female
- Other:

Branch *

Another Individual Response



What do you typically do during gap hours? (Select all that apply)

- [x] Self-study
- [ ] Socializing with friends
- [x] Scrolling phone/social media
- [x] Sleeping/resting,
- [ ] Eating/snacking

Productivity & Fatigue Impact

On a scale of 1-10, how productive are your gap hours? (1 = Not at all, 10 = Very productive) *

1 2 3 4 5 6 7 8 9 10

How many hours across these 2 busiest days do you feel were "wasted" (neither restful nor productive)? Enter a number *

3

Rate your physical fatigue level at 6 PM on your busiest day (1 = Energetic, 10 = Completely drained) *

1 2 3 4 5 6 7 8 9 10

Do you feel the gaps in your timetable significantly decrease your overall academic focus? *

- ( ) Yes
- ( ) No
- (•) Maybe

Would you prefer a more compact/continuous class schedule? *

- ( ) strongly agree
- (•) agree
- ( ) neutral
- ( ) disagree
- ( ) strongly disagree

Examples of Response Received